YUANCHENG XU

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EDUCATION

University of Maryland, College Park *2020-2025*
Ph.D. in Applied Mathematics and Statistics, and Scientific Computation
Advisor: Furong Huang

Southern University of Science and Technology, China *2016-2020*
B.S. in Mathematics and Applied Mathematics
GPA 3.94/4.00 (1/909), Summa Cum Laude (1%)

RESEARCH INTERESTS

AI Safety and Alignment; Reasoning and Reinforcement Learning; Multimodal Understanding and Generation

WORK EXPERIENCE

Netflix
Senior Research Scientist Sept 2026 - Present
Research Scientist July 2025 - Sept 2026

- **Video Foundation Model Post-training:** Drive large-scale post-training of multimodal video foundation models, advancing cross-modal reasoning and instruction following for fine-grained, controllable video generation aligned with complex creative intent.
- **Learning Formulation:** Pioneered a novel learning formulation that overcame the lack of paired supervision for identity- and performance-preserving video editing, enabling complex visual transformations while retaining fine-grained source details and motion.
- **Data-centric Research:** Designed multimodal reasoning data to improve cross-modal grounding and compositional instruction following; developed scalable, geometry-aware curation methods to transform noisy production footage into high-quality training data.
- **Production Impact:** Delivered production-grade 4K video generation capabilities with precise creative control and high reference fidelity, adopted across multiple Netflix filmmaking workflows.

Netflix Eyeline Labs Sept 2024 - March 2025
Research Intern
- Developed video generation model with motion control and 3D consistency.

JPMorgan Chase & Co. June 2024 - Aug 2024
AI Research Intern
- Developed an Autoregressive Reward Model for test-time alignment of large language models.

SELECTED PUBLICATIONS

Please visit my Google Scholar for the complete list.

Safety and Alignment

1. PropensityBench: Evaluating Latent Safety Risks in Large Language Models via an Agentic Approach.
U. M. Sehwag, S. Shabihi, A. McAvoy, V. Sehwag, **Y. Xu**, D. Towers, F. Huang. *ICLR 2026.*
2. PARM: Multi-objective Test-time Alignment via Preference-aware Autoregressive Reward Model.
B. Lin, W. Jiang, **Y. Xu**, H. Chen, Y.-C. Chen. *ICML 2025.*
3. GenARM: Reward Guided Generation with Autoregressive Reward Model for Test-Time Alignment.
Y. Xu, U. M. Sehwag, A. Koppel, S. Zhu, B. An, F. Huang, S. Ganesh. *ICLR 2025.*

4. Can Watermarking Large Language Models Prevent Copyrighted Text Generation and Hide Training Data?
M.-A. Panaitescu-Liess, Z. Che, B. An, **Y. Xu**, P. Pathmanathan, S. Chakraborty, S. Zhu, T. Goldstein, F. Huang. *AAAI 2025.*
New Frontiers in Adversarial Machine Learning Workshop at NeurIPS 2024 (Best Paper Award).

5. Shadowcast: Stealthy Data Poisoning Attacks Against Vision-Language Models.
Y. Xu, J. Yao, M. Shu, Y. Sun, Z. Wu, N. Yu, T. Goldstein, F. Huang. *NeurIPS 2024.*

6. Automatic Pseudo-Harmful Prompt Generation for Evaluating False Refusals in Large Language Models.
B. An, S. Zhu, R. Zhang, M.-A. Panaitescu-Liess, **Y. Xu**, F. Huang. *COLM 2024.*

7. Adapting Static Fairness to Sequential Decision-Making: Bias Mitigation Strategies towards Equal Long-term Benefit Rate.
Y. Xu, C. Deng, Y. Sun, R. Zheng, X. Wang, J. Zhao, F. Huang. *ICML 2024.*

8. WAVES: Benchmarking the Robustness of Image Watermarks.
B. An, M. Ding, T. Rabbani, A. Agrawal, **Y. Xu**, C. Deng, S. Zhu, A. Mohamed, Y. Wen, T. Goldstein, F. Huang. *ICML 2024.*

9. Exploring and Exploiting Decision Boundary Dynamics for Adversarial Robustness.
Y. Xu, Y. Sun, M. Goldblum, T. Goldstein, F. Huang. *ICLR 2023.*

Video Generation

10. ID-V2V: Identity-preserving Video Restylization.
Y. Xu, M. He, P. Salamanca, L. Ma, Y. Kant, E. Steven, P. Debevec, N. Yu. *SIGGRAPH Asia 2026.*

11. Vista4D: Video Reshooting with 4D Point Clouds.
K. H. Lin, Z. Liu, P. Salamanca, Y. Kant, R. Burgert, **Y. Xu**, K. Namekata, Y. Zhao, B. Zhou, M. Goldblum, P. Debevec, N. Yu. *CVPR 2026 (Highlight).*

12. Virtually Being: Customizing Camera-Controllable Video Diffusion Models with Volumetric Performance Captures.
Y. Xu, W. Xian, L. Ma, J. Philip, A. Taşel, Y. Zhao, R. Burgert, M. He, O. Hermann, O. Pilarski, R. Garg, P. Debevec, N. Yu. *SIGGRAPH Asia 2025.*

13. Go-with-the-Flow: Motion-Controllable Video Diffusion Models Using Real-Time Warped Noise.
R. Burgert, **Y. Xu**, W. Xian, O. Pilarski, P. Clausen, M. He, L. Ma, Y. Deng, L. Li, M. Mousavi, M. Ryoo, P. Debevec, N. Yu. *CVPR 2025 (Oral Presentation).*

Multimodal Reasoning

14. Scaling Inference-time Search with Vision Value Model for Improved Visual Comprehension.
X. Wang, Z. Yang, L. Li, H. Lu, **Y. Xu**, C.-C. Lin, K. Lin, F. Huang, L. Wang. *ICCV 2025.*

15. Mementos: A Comprehensive Benchmark for Multimodal Large Language Model Reasoning over Image Sequences.
X. Wang, Y. Zhou, X. Liu, H. Lu, **Y. Xu**, F. He, J. Yoon, T. Lu, F. Liu, G. Bertasius, M. Bansal, H. Yao, F. Huang. *ACL 2024.*

AWARDS

Dean's fellowship, University of Maryland, College Park	2020,2021
Summa Cum Laude at Southern University of Science and Technology (10/1000)	2020
China National Scholarship (0.2%, Highest honor of Chinese undergraduate students)	2019